

Zhengyuan Yang

Principal Researcher, Microsoft, Redmond, WA

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Research Interests

My research centers on multimodal foundation models and post-train them for solving long-horizon tasks. Most recently, I have been building large-scale multimodal reasoning and agentic models. Earlier, I led the diffusion distillation for DALL-E 3, and conducted research on vision-language understanding and generation.

Education

University of Rochester

Rochester, New York

PH.D. IN COMPUTER SCIENCE ADVISOR: PROF. JIEBO LUO

2016 - 2021

- Thesis: Visual Grounding: Building Cross-Modal Visual-Text Alignment (**ACM SIGMM Outstanding Ph.D. Thesis**)

University of Science and Technology of China

Hefei, China

B.E. IN ELECTRONIC ENGINEERING AND INFORMATION SCIENCE

2012 - 2016

Experience

Microsoft

Redmond, WA

PRINCIPAL RESEARCHER

Sept. 2024 - Current

SENIOR RESEARCHER

Sept. 2022 - Sept. 2024

RESEARCHER

June 2021 - Sept. 2022

- Training large-scale multimodal reasoning and agentic models for long-horizon tasks.
- Long-trajectory RL post-train of Operator (computer using agent) and DeepResearch.
- Lead DALL-E 3 diffusion distillation.
- Research on scalable multimodal understanding and generation.

Microsoft

Bellevue, WA

RESEARCH INTERN

SUPERVISOR: DR. YIJUAN LU, DR. JIANFENG WANG, DR. XI YIN

May 2020 to Aug 2020

Tencent AI Lab at Bellevue

Bellevue, WA

RESEARCH INTERN

SUPERVISOR: DR. BOQING GONG, DR. LIWEI WANG

Jan 2019 to Apr 2019

SnapChat Research

Venice, CA

RESEARCH INTERN

SUPERVISOR: DR. YUNCHENG LI, DR. LINJIE YANG, DR. NING ZHANG

May 2018 to Aug 2018

Selected Publications

NeurIPS 2025a

Xiyao Wang*, **Zhengyuan Yang***, Chao Feng, Yongyuan Liang, Yuhang Zhou, Xiaoyu Liu, Ziyi Zang, Ming Li, Chung-Ching Lin, Kevin Lin, Linjie Li, Furong Huang, Lijuan Wang, "ViCrit: A Verifiable Reinforcement Learning Proxy Task for Visual Perception in VLMs," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.

NeurIPS 2025b

Xiyao Wang, **Zhengyuan Yang**, Chao Feng, Hongjin Lu, Linjie Li, Chung-Ching Lin, Kevin Lin, Furong Huang, Lijuan Wang, "SoTA with Less: MCTS-Guided Sample Selection for Data-Efficient Visual Reasoning Self-Improvement," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025. **Spotlight presentation.**

NeurIPS 2025c

Minheng Ni, **Zhengyuan Yang**, Linjie Li, Chung-Ching Lin, Kevin Lin, Wangmeng Zuo, Lijuan Wang, "Point-RFT: Improving Multimodal Reasoning with Visually Grounded Reinforcement Finetuning," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.

NeurIPS 2025d

Kangrui Wang, Pingyue Zhang, Zihan Wang, Yaning Gao, Linjie Li, Qineng Wang, Hanyang Chen, Yiping Lu, **Zhengyuan Yang**, Lijuan Wang, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Yejin Choi, Manling Li, "Reinforcing Visual State Reasoning for Multi-Turn VLM Agents," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.

NeurIPS 2025e	Jitesh Jain, Zhengyuan Yang , Humphrey Shi, Jianfeng Gao, Jianwei Yang, "Elevating Visual Perception in Multimodal LLMs with Visual Embedding Distillation," The 39th Conference on Neural Information Processing Systems (NeurIPS), San Diego, CA, December 2025.
EMNLP 2025	Yiyang Zhou, Linjie Li, Shi Qiu, Zhengyuan Yang , Yuyang Zhao, Siwei Han, Yangfan He, Kangqi Li, Haonian Ji, Zihao Zhao, Haibo Tong, Lijuan Wang, Huaxiu Yao, "GLIMPSE: LLMs Can Watch but Do They Truly Understand?," The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), Suzhou, China, Nov 2025.
ICCV 2025a	Xiyao Wang, Zhengyuan Yang , Linjie Li, Hongjin Lu, Yuancheng Xu, Chung-Ching Lin, Kevin Lin, Furong Huang, Lijuan Wang, "Scaling Inference-time Search with Vision Value Model for Improved Visual Comprehension," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.
ICCV 2025b	Jiaqi Liao, Zhengyuan Yang , Linjie Li, Dianqi Li, Kevin Lin, Yu Cheng, Lijuan Wang, "ImageGen-CoT: Enhancing Text-to-Image In-context Learning with Chain-of-Thought Reasoning," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.
ICCV 2025c	Wenqi Wang, Reuben Tan, Pengyue Zhu, Jianwei Yang, Zhengyuan Yang , Lijuan Wang, Andrey Kolobov, Jianfeng Gao, Boqing Gong, "SITE: towards Spatial Intelligence Thorough Evaluation," International Conference on Computer Vision (ICCV), Honolulu, Hawaii, Oct 2025.
ICML 2025a	Xingyu Fu, Minqian Liu, Zhengyuan Yang , John Corring, Yijuan Lu, Jianwei Yang, Dan Roth, Dinei Florencio, Cha Zhang, "ReFocus: Visual Editing as a Chain of Thought for Structured Image Understanding," The 42nd International Conference on Machine Learning (ICML), Vancouver, Canada, July 2025.
ICML 2025b	Yunzhuo Hao, Jiawei Gu, Huichen Will Wang, Linjie Li, Zhengyuan Yang , Lijuan Wang, Yu Cheng, "Can MLLMs Reason/Think in Multimodality? EMMA: An Enhanced MultiModal Reasoning Benchmark," The 42nd International Conference on Machine Learning (ICML), Vancouver, July 2025. Oral presentation (1.0%).
CVPR 2025a	Kevin Qinghong Lin, Linjie Li, Difei Gao, Zhengyuan Yang , Shiwei Wu, Zechen Bai, Weixian Lei, Lijuan Wang, Mike Zheng Shou, "ShowUI: One Vision-Language-Action Model for GUI Visual Agent," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, June 2025.
CVPR 2025b	Qin Liu, Jianfeng Wang, Zhengyuan Yang , Linjie Li, Kevin Lin, Marc Niethammer, Lijuan Wang, "LiVOS: Light Video Object Segmentation with Gated Linear Matching," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, June 2025.
NAACL 2025	Chenglei Si, Yanzhe Zhang, Ryan Li, Zhengyuan Yang , Ruibo Liu, Diyi Yang, "Design2Code: Benchmarking Multimodal Code Generation for Automated Front-End Engineering," The Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), New Mexico, April 2025. Oral presentation.
ICLR 2025a	Yining Hong, Beide Liu, Maxine Wu, Yuanhao Zhai, Kai-Wei Chang, Linjie Li, Kevin Lin, Chung-Ching Lin, Jianfeng Wang, Zhengyuan Yang* , Yingnian Wu*, Lijuan Wang*, "SlowFast-VGen: Slow-Fast Learning for Action-Driven Long Video Generation," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025. Spotlight presentation.
ICLR 2025b	Zichen Miao, Zhengyuan Yang , Kevin Lin, Ze Wang, Zicheng Liu, Lijuan Wang, Qiang Qiu, "Tuning Timestep-Distilled Diffusion Model Using Pairwise Sample Optimization," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.
ICLR 2025c	Yuyang Zhao, Chung-Ching Lin, Kevin Lin, Zhiwen Yan, Linjie Li, Zhengyuan Yang , Jianfeng Wang, Gim Hee Lee, Lijuan Wang, "GenXD: Generating Any 3D and 4D Scenes," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.
ICLR 2025d	Xuehai He, Weixi Feng, Kaizhi Zheng, Yujie Lu, Wanrong Zhu, Jiachen Li, Yue Fan, Jianfeng Wang, Linjie Li, Zhengyuan Yang , Kevin Lin, William Yang Wang, Lijuan Wang, Xin Eric Wang, Zhengyuan Yang , Jianfeng Wang, Gim Hee Lee, Lijuan Wang, "MMWorld: Towards Multi-discipline Multi-faceted World Model Evaluation in Videos," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.
ICLR 2025e	Kaizhi Zheng, Xiaotong Chen, Xuehai He, Jing Gu, Linjie Li, Zhengyuan Yang , Kevin Lin, Jianfeng Wang, Lijuan Wang, Xin Eric Wang, "EditRoom: LLM-parameterized Graph Diffusion for Composable 3D Room Layout Editing," The Thirteenth International Conference on Learning Representations (ICLR), Singapore, April 2025.

NeurIPS 2024a	Yuanhao Zhai, Kevin Lin, Zhengyuan Yang , Linjie Li, Jianfeng Wang, Chung-Ching Lin, David Doermann, Junsong Yuan, Lijuan Wang, "Motion Consistency Model: Accelerating Video Diffusion with Disentangled Motion-Appearance Distillation," The 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, BC, December 2024.
NeurIPS 2024b	Xueyan Zou, Linjie Li, Jianfeng Wang, Jianwei Yang, Mingyu Ding, Junyi Wei, Zhengyuan Yang , Feng Li, Hao Zhang, Shilong Liu, Arul Aravinthan, Yong Jae Lee, Lijuan Wang, "Interfacing Foundation Models' Embeddings," The 38th Conference on Neural Information Processing Systems (NeurIPS), Vancouver, BC, December 2024.
NeurIPS 2024c	Kevin Qinghong Lin, Linjie Li, Difei Gao, Qinchen WU, Mingyi Yan, Zhengyuan Yang , Lijuan Wang, Mike Zheng Shou, "VideoGUI: A Benchmark for GUI Automation from Instructional Videos," The 38th Conference on Neural Information Processing Systems Datasets and Benchmarks (NeurIPS), Vancouver, BC, December 2024. Spotlight presentation.
COLM 2024	An Yan, Zhengyuan Yang , Junda Wu, Wanrong Zhu, Jianwei Yang, Linjie Li, Kevin Lin, Jianfeng Wang, Julian McAuley, Jianfeng Gao, Lijuan Wang, "List Items One by One: A New Data Source and Learning Paradigm for Multimodal LLMs," The 1st Conference on Language Modeling (COLM), Philadelphia, PA, October 2024.
ACMMM 2024	Jie An, Zhengyuan Yang , Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Lijuan Wang, Jiebo Luo, "Openleaf: Open-domain interleaved image-text generation and evaluation," ACM Multimedia Conference (MM), Brave New Ideas track, Melbourne, Australia, October 2024. Oral presentation.
ECCV 2024a	Zhengyuan Yang , Jianfeng Wang, Linjie Li, Kevin Lin, Chung-Ching Lin, Zicheng Liu, Lijuan Wang, "Idea2Img: Iterative Self-Refinement with GPT-4V(ision) for Automatic Image Design and Generation," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.
ECCV 2024b	Jialian Wu, Jianfeng Wang, Zhengyuan Yang , Zhe Gan, Zicheng Liu, Junsong Yuan, Lijuan Wang, "GRiT: A Generative Region-to-text Transformer for Object Understanding," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.
ECCV 2024c	Yuanhao Zhai, Kevin Lin, Linjie Li, Chung-Ching Lin, Jianfeng Wang, Zhengyuan Yang , David Doermann, Junsong Yuan, Zicheng Liu, Lijuan Wang, "IDOL: Unified Dual-Modal Latent Diffusion for Human-Centric Joint Video-Depth Generation," The 18th European Conference on Computer Vision (ECCV), Milano, Italy, Sept 2024.
ICML 2024a	Weihaoyu*, Zhengyuan Yang *, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, Lijuan Wang, "MM-Vet: Evaluating Large Multimodal Models for Integrated Capabilities," The Forty-first International Conference on Machine Learning (ICML), Vienna, Austria, July 2024.
ICML 2024b	Zecheng Tang, Chenfei Wu, Zekai Zhang, Mingheng Ni, Shengming Yin, Yu Liu, Zhengyuan Yang , Lijuan Wang, Zicheng Liu, Juntao Li, Nan Duan, "StrokeNUWA: Tokenizing Strokes for Vector Graphic Synthesis," The Forty-first International Conference on Machine Learning (ICML), Vienna, Austria, July 2024.
IJCAI 2024	Jie An, Zhengyuan Yang , Jianfeng Wang, Linjie Li, Zicheng Liu, Lijuan Wang, Jiebo Luo, "Bring Metric Functions into Diffusion Models," The 32nd International Joint Conference on Artificial Intelligence (IJCAI), Jeju, August 2024.
CVPR 2024a	Chaoyi Zhang, Kevin Lin, Zhengyuan Yang , Jianfeng Wang, Linjie Li, Chung-Ching Lin, Zicheng Liu, Lijuan Wang, "MM-Narrator: Narrating Long-form Videos with Multimodal In-Context Learning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024. Highlight presentation.
CVPR 2024b	Tan Wang, Linjie Li, Kevin Lin, Yuanhao Zhai, Chung-Ching Lin, Zhengyuan Yang , Hanwang Zhang, Zicheng Liu, Lijuan Wang, "DisCo: Disentangled Control for Realistic Human Dance Generation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024.
CVPR 2024c	Zichen Miao, Jiang Wang, Ze Wang, Zhengyuan Yang , Lijuan Wang, Qiang Qiu, Zicheng Liu, "Training Diffusion Models Towards Diverse Image Generation with Reinforcement Learning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024.
CVPR 2024d	Jielin Qiu, Jiacheng Zhu, William Han, Aditesh Kumar, Karthik Mittal, Claire Jin, Zhengyuan Yang , Linjie Li, Jianfeng Wang, Ding Zhao, Bo Li, Lijuan Wang, "MMSum: A Dataset for Multimodal Summarization and Thumbnail Generation of Videos," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Seattle, WA, June 2024. Highlight presentation.

Book 2023	Chunyu Li*, Zhe Gan*, Zhengyuan Yang* , Jianwei Yang*, Linjie Li*, Lijuan Wang, Jianfeng Gao, "Multimodal Foundation Models: From Specialists to General-Purpose Assistants," Foundations and Trends in Computer Graphics and Vision, 2023.
Arxiv 2023a	Zhengyuan Yang* , Linjie Li*, Kevin Lin*, Jianfeng Wang*, Chung-Ching Lin*, Zicheng Liu, Lijuan Wang*, "The Dawn of LMMs: Preliminary Explorations with GPT-4V(ision)." (Exploratory work cataloguing use of GPT-4V).
Arxiv 2023b	Zhengyuan Yang , Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, Lijuan Wang, "MM-REACT: Prompting ChatGPT for Multimodal Reasoning and Action."
ICCV 2023a	Yushi Hu, Hang Hua, Zhengyuan Yang , Weijia Shi, Noah A. Smith, Jiebo Luo, "PromptCap: Prompt-Guided Task-Aware Image Captioning," IEEE International Conference on Computer Vision (ICCV), Paris, France, October 2023.
ICCV 2023b	Tan Wang, Kevin Lin, Linjie Li, Chung-Ching Lin, Zhengyuan Yang , Hanwang Zhang, Zicheng Liu, Lijuan Wang, "Equivariant Similarity for Vision-Language Foundation Models," IEEE International Conference on Computer Vision (ICCV), Paris, France, October 2023. Oral presentation .
ACL 2023	Shengming Yin, Chenfei Wu, Huan Yang, Jianfeng Wang, Xiaodong Wang, Minheng Ni, Zhengyuan Yang , Linjie Li, Shuguang Liu, Fan Yang, Jianlong Fu, Gong Ming, Lijuan Wang, Zicheng Liu, Houqiang Li, Nan Duan, "NUWA-XL: Diffusion over Diffusion for eXtremely Long Video Generation." Annual Meeting of the Association for Computational Linguistics (ACL), Toronto, Canada, July 2023. Oral presentation .
IJCAI 2023	Xiaodong Wang, Chenfei Wu, Shengming Yin, Minheng Ni, Jianfeng Wang, Linjie Li, Zhengyuan Yang , Fan Yang, Lijuan Wang, Zicheng Liu, Yuejian Fang, Nan Duan, "Learning 3D Photography Videos via Self-supervised Diffusion on Single Images." The 32nd International Joint Conference on Artificial Intelligence (IJCAI), Macao, August 2023.
CVPR 2023	Zhengyuan Yang , Jianfeng Wang, Zhe Gan, Linjie Li, Kevin Lin, Chenfei Wu, Nan Duan, Zicheng Liu, Ce Liu, Michael Zeng, Lijuan Wang, "ReCo: Region-Controlled Text-to-Image Generation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Vancouver, BC, June 2023.
ICLR 2023	Chenglei Si, Zhe Gan, Zhengyuan Yang , Shuohang Wang, Jianfeng Wang, Jordan Boyd-Graber, Lijuan Wang, "Prompting GPT-3 To Be Reliable," The Eleventh International Conference on Learning Representations (ICLR), Kigali, Rwanda, May 2023.
ECCV 2022	Zhengyuan Yang , Zhe Gan, Jianfeng Wang, Xiaowei Hu, Faisal Ahmed, Zicheng Liu, Yumao Lu, Lijuan Wang, "UniTAB: Unifying Text and Box Outputs for Grounded Vision-Language Modeling," European Conference on Computer Vision (ECCV), Tel Aviv, Israel, October 2022. Oral presentation (2.7%) .
TPAMI 2023	Jiajun Deng, Zhengyuan Yang , Daqing Liu, Tianlang Chen, Wengang Zhou, Yanyong Zhang, Houqiang Li, Wanli Ouyang, "TransVG++: End-to-End Visual Grounding with Language Conditioned Vision Transformer," IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2023.
TMLR 2022	Jianfeng Wang, Zhengyuan Yang , Xiaowei Hu, Linjie Li, Kevin Lin, Zhe Gan, Zicheng Liu, Ce Liu, Lijuan Wang, "GIT: A Generative Image-to-text Transformer for Vision and Language," Transactions on Machine Learning Research (TMLR), 2022.
CVPR 2022	Xiaowei Hu, Zhe Gan, Jianfeng Wang, Zhengyuan Yang , Zicheng Liu, Yumao Lu and Lijuan Wang, "Scaling Up Vision-Language Pre-training for Image Captioning," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, June 2022.
Arxiv 2021	Jianfeng Wang, Xiaowei Hu, Zhe Gan, Zhengyuan Yang , Xiyang Dai, Zicheng Liu, Yumao Lu and Lijuan Wang, "UFO: A UniFied TransFormer for Vision-Language Representation Learning."
AAAI 2022	Zhengyuan Yang , Zhe Gan, Jianfeng Wang, Xiaowei Hu, Yumao Lu, Zicheng Liu, Lijuan Wang, "An Empirical Study of GPT-3 for Few-Shot Knowledge-Based VQA," The 36th AAAI Conference on Artificial Intelligence (AAAI), Vancouver, BC, February 2022. Oral presentation (4.2%) .
ICPR 2022	Zhengyuan Yang , Jingen Liu, Jing Huang, Xiaodong He, Tao Mei, Chenliang Xu, Jiebo Luo, "Cross-modal Contrastive Distillation for Instructional Activity Anticipation," International Conference on Pattern Recognition (ICPR), Montreal, Quebec, Canada, August 2022.
ICCV 2021a	Zhengyuan Yang , Songyang Zhang, Liwei Wang, Jiebo Luo, "SAT: 2D Semantics Assisted Training for 3D Visual Grounding," IEEE International Conference on Computer Vision (ICCV), Online, 2021. Oral presentation (3.4%) .

ICCV 2021b	Jiajun Deng, Zhengyuan Yang , Tianlang Chen, Wengang Zhou, Houqiang Li, "TransVG: End-to-End Visual Grounding with Transformers," IEEE International Conference on Computer Vision (ICCV), Online, 2021.
CVPR 2021a	Zhengyuan Yang , Yijuan Lu, Jianfeng Wang, Xi Yin, Dinei Florencio, Lijuan Wang, Cha Zhang, Lei Zhang, Jiebo Luo, "TAP: Text-Aware Pre-training for Text-VQA and Text-Caption," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Online, June 2021. Oral presentation (4.0%).
CVPR 2021b	Liwei Wang, Jing Huang, Yin Li, Kun Xu, Zhengyuan Yang , Dong Yu, "Improving Weakly Supervised Visual Grounding by Contrastive Knowledge Distillation," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Online, June 2021.
ECCV 2020	Zhengyuan Yang , Tianlang Chen, Liwei Wang, Jiebo Luo, "Improving One-stage Visual Grounding by Recursive Sub-query Construction," European Conference on Computer Vision (ECCV), Online, August 2020.
ACL 2020	Yongjing Yin, Fandong Meng, Jinsong Su, Chulun Zhou, Zhengyuan Yang , Jie Zhou, Jiebo Luo, "A Novel Graph-based Multi-modal Fusion Encoder for Neural Machine Translation," Annual Meeting of the Association for Computational Linguistics (ACL), Online, July 2020.
T-CSVT 2020	Zhengyuan Yang , Tushar Kumar, Tianlang Chen, Jingsong Su, Jiebo Luo, "Grounding-Tracking-Integration," IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT).
ACMMM 2020	Huan Lin, Fandong Meng, Jinsong Su, Yongjing Yin, Zhengyuan Yang , Yubin Ge, Jie Zhou, Jiebo Luo, "Dynamic Context-guided Capsule Network for Multimodal Machine Translation," ACM Multimedia Conference, Seattle, WA, October 2020
ICCV 2019	Zhengyuan Yang , Boqing Gong, Liwei Wang, Wenbing Huang, Dong Yu, Jiebo Luo, "A Fast and Accurate One-Stage Approach to Visual Grounding," IEEE International Conference on Computer Vision (ICCV), Seoul, South Korea, 2019. Oral presentation (4.3%).
CVPR 2019	Mengshi Qi, Weijian Li, Zhengyuan Yang , Yunhong Wang, Jiebo Luo, "Attentive Relational Networks for Mapping Images to Scene Graphs," IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Long Beach, USA, 2019.
T-CSVT 2018	Zhengyuan Yang , Yuncheng Li, Jianchao Yang, Jiebo Luo, "Action Recognition with Spatio-Temporal Visual Attention on Skeleton Image Sequences," IEEE Transactions on Circuits and Systems for Video Technology (T-CSVT).
ICPR 2018a	Zhengyuan Yang , Yixuan Zhang, Jerry Yu, Junjie Cai, Jiebo Luo, "End-to-end Multi-Modal Multi-Task Vehicle Control for Self-Driving Cars with Visual Perceptions," International Conference on Pattern Recognition (ICPR), Beijing, China, 2018. Best Industry Related Paper Award (BIRPA) (1/1258).

Talks

Tutorial Talk – Multimodal Reasoning for Visual-Centric Long-Horizon Tasks	<i>Nashville, TN</i>
• CVPR 2025 Tutorial on Recent Advances in Vision Foundation Models	2025
Invited Talk – Benchmarks in the Era of Multimodal Reasoning and Agentic Models	<i>Nashville, TN</i>
• CVPR 2025 Workshop on Computer Vision in the Wild	2025
Invited Talk – Multimodal Agents	<i>Niagara Falls, Canada</i>
• ICME 2024 Workshop on Visual-Language Alignment in Text-Guided Multi-Modal Generation	2024
Tutorial Talk – Recent Advances in Image Generative Foundation Models	<i>Seattle, WA</i>
• CVPR 2024 Tutorial on Recent Advances in Vision Foundation Models	2024
Invited Talk – Multimodal Agents: from Text to Multimodal Reasoning and Action	<i>Hong Kong, China</i>
• CUHK Multi-Modal Symposium	2024
Tutorial Talk – Alignments in Text-to-Image Generation	<i>Vancouver, BC</i>
• CVPR 2023 Tutorial on Recent Advances in Vision Foundation Models	2023
Invited Talk – Towards Cross-Modal Visual-Text Understanding and Generation	<i>Tokyo, Japan</i>
• ACM MM Asia 2022 Workshop on Multimedia Understanding with Pre-trained Models	2022

Tutorial Talk – Unified Image-Text Modeling

- CVPR 2022 Tutorial on Recent Advances in Vision-and-Language Pre-training

New Orleans, LA

2022

Invited Talk – SAT: 2D Semantics Assisted Training for 3D Visual Grounding

- CVPR Workshop 2021 on Language for 3D Scenes

Online

2021

Guest Lecture – Vision-and-language

- CS440 Data Mining, University of Rochester

Rochester, NY

2021

Guest Lecture – CNN and Feature Visualization

- CS440 Data Mining, University of Rochester

Rochester, NY

2020

Selected Awards

ACM SIGMM Award for Outstanding Ph.D. Thesis

2022

ECCV 2022 Outstanding Reviewer

2022

Winner of CVPR 2021 TextCaps Challenge

2021

Winner of CVPR 2021 ReferIt3D Challenge

2021

CVPR 2021 Outstanding Reviewer

2021

Twitch Research Fellowship

2020

Best Industry Related Paper Award (BIRPA) in ICPR 2018, (1/1258)

2018

Services

Exhibits and Demos Chair: IEEE International Conference on Multimedia and Expo (ICME)

2024

Area Chair: ICLR, EMNLP, ACMMM, NAACL

Senior Program Committee: AAAI Conference on Artificial Intelligence (AAAI)

2023-2026

Associate Editor: IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)

2022-2026

Guest Editor: IEEE TMM SI on “Large Multimodal Models for Dynamic Visual Scene Understanding”

2024

IEEE TCSVT SI on “AI-Generated Content for Multimedia”

2023

Conference Reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, ICML, ACL, EMNLP, ACMMM, AAAI, ACCV, WACV, ICME, ICIP.

Journal Reviewer: TPAMI, IJCV, TIP, TMM, TCybernetics, TCSVT, Pattern Recognition, Neurocomputing, TBioCAS, Access.

Tutorial Organizer: Recent Advances in Vision Foundation Models at CVPR

2022-2025

Tutorial Organizer: Computer Vision in the Wild at CVPR

2025

Selected Patents

“Weakly supervised semantic parsing”, *US Patent Number: 11,182,603* (Granted, 11/2021).